MINI PROJECT

Spam Mail Detection Using Artificial Intelligence and Machine Learning

ABSTRACT

Spam emails are unwanted messages sent in bulk for advertising, phishing, fraud, or malicious purposes. With the rapid growth of internet communication, spam emails have become a major problem for users and organizations.

This project develops a Spam Mail Detection System using Artificial Intelligence and Machine Learning techniques. The system uses a Logistic Regression algorithm along with TF-IDF (Term Frequency-Inverse Document Frequency) vectorization to classify emails as either Spam or Ham (Legitimate Email).

The developed system accepts email content from users and predicts whether the message is spam or genuine. The project is deployed as a web application using Django, making it accessible through a browser.

INTRODUCTION

Email is one of the most commonly used communication methods worldwide. However, a large number of emails received daily are spam messages.

These spam emails may contain:

- Advertisements
- Fraudulent offers
- Phishing links
- Malware
- Fake lottery messages

Manual identification of spam emails is difficult and time-consuming.

Machine Learning can automatically learn patterns from previous emails and classify new emails efficiently.

This project aims to create an intelligent system that automatically identifies spam emails.

PROBLEM STATEMENT

Traditional filtering methods based on fixed rules cannot accurately detect all spam emails.

The objective is to develop a Machine Learning-based system that:

- Analyzes email content.
- Detects spam emails automatically.
- Reduces user effort.
- Improves email security.

OBJECTIVES

The primary objectives of this project are:

1. To classify emails into Spam and Ham categories.
2. To use Machine Learning techniques for prediction.
3. To provide a user-friendly web interface.
4. To improve email security.
5. To achieve high prediction accuracy.

SCOPE OF THE PROJECT

The system can be used in:

- Educational Institutions
- Corporate Organizations
- Email Service Providers
- Personal Email Management
- Cybersecurity Applications

Future enhancements can include:

- Real-time Gmail Integration
- Multilingual Email Detection
- Deep Learning Models
- Attachment Analysis
- Phishing Detection

Technology Stack

Component	**Technology**
Programming Language	Python
Machine Learning	Scikit-Learn
Data Handling	Pandas, NumPy
Web Framework	Django
Frontend	HTML, CSS
Dataset	Email Spam Dataset
Model	Logistic Regression
Feature Extraction	TF-IDF Vectorizer
Deployment	Render

SYSTEM ARCHITECHTURE

User Input

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Email Content

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TF-IDF Vectorizer

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Logistic Regression Model

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Spam / Ham Prediction

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Display Result

DATASET DESCRIPTION

The dataset contains two columns:

Column	Description
Category	Spam or Ham
Message	Email Content

Example:

Category	Message
spam	Win ₹50000 now!
ham	Meeting at 10 AM tomorrow

METHODOLOGY

Step 1: Data Collection

The email dataset is loaded using Pandas.

```
raw_mail_data = pd.read_csv("mail_data.csv")
```

Step 2: Data Preprocessing

Missing values are replaced with empty strings.

```
mail_data = raw_mail_data.where(
   (pd.notnull(raw_mail_data)),
   ''
)
```

Step 3: Label Encoding

Spam emails are converted to 0.

Ham emails are converted to 1.

spam → 0
ham → 1

Step 4: Train-Test Split

Dataset is divided into:

- Training Data (80%)
- Testing Data (20%)

```
train_test_split(
   x,
   y,
```

```
    test_size=0.2
)
```

Step 5: Feature Extraction

TF-IDF converts text into numerical vectors.

Example:

"free"
"winner"
"offer"

become machine-readable numerical values.

TfidfVectorizer()

Step 6: Model Training

Logistic Regression algorithm is used.

```
model.fit(
    x_train_features,
    y_train
)
```

The model learns patterns from spam and ham emails.

Step 7: Prediction

New emails are converted into vectors and classified.

```
prediction =
model.predict(
    input_data_features
)
```

11. Machine Learning Algorithm Used

Logistic Regression

Logistic Regression is a supervised learning classification algorithm.

Advantages

- Fast Training
- High Accuracy
- Simple Implementation
- Good for Text Classification

Why Logistic Regression?

- Works efficiently on text data.
- Suitable for binary classification.
- Requires less computational power.

12. TF-IDF Vectorization

TF-IDF stands for:

Term Frequency – Inverse Document Frequency

It converts text into numerical form.

Example:

Congratulations
Winner
Prize
Offer

These words receive different weights based on importance.

Benefits:

- Removes irrelevant words.
- Improves prediction accuracy.
- Converts text into machine-readable format.

MODEL EVALUATION

Accuracy is calculated using:

accuracy_score()

Training Accuracy:

≈ 96% - 99%

Testing Accuracy:

≈ 95% - 98%

(Actual values depend on dataset splitting.)

USER INTERFACE

The web application allows users to:

- Paste Email Content
- Analyze Emails
- View Spam/Ham Results

Features:

- Responsive Design
- Mobile Friendly
- Modern User Interface

Advantages

- Fast Detection
- User Friendly
- Accurate Predictions
- Low Resource Consumption
- Easy Deployment
- Scalable

Limitations

- Depends on training dataset quality.
- May misclassify very new spam patterns.
- Does not analyze attachments.
- Does not detect advanced phishing attacks.

Future Enhancements

1. Gmail API Integration
2. Outlook Integration
3. Deep Learning Models
4. BERT-Based Spam Detection
5. Attachment Analysis
6. Real-Time Email Monitoring
7. Multi-Language Spam Detection
8. Phishing URL Detection

CONCLUSION

The Spam Mail Detection System using Artificial Intelligence and Machine Learning successfully classifies emails into Spam and Ham categories. The project uses TF-IDF vectorization and Logistic Regression to achieve high accuracy in email classification.

The developed Django web application provides an easy-to-use interface where users can submit email content and receive instant predictions. This system contributes to improving email security and reducing the impact of spam communication.